Trefoil Symplectic-Period Certificate

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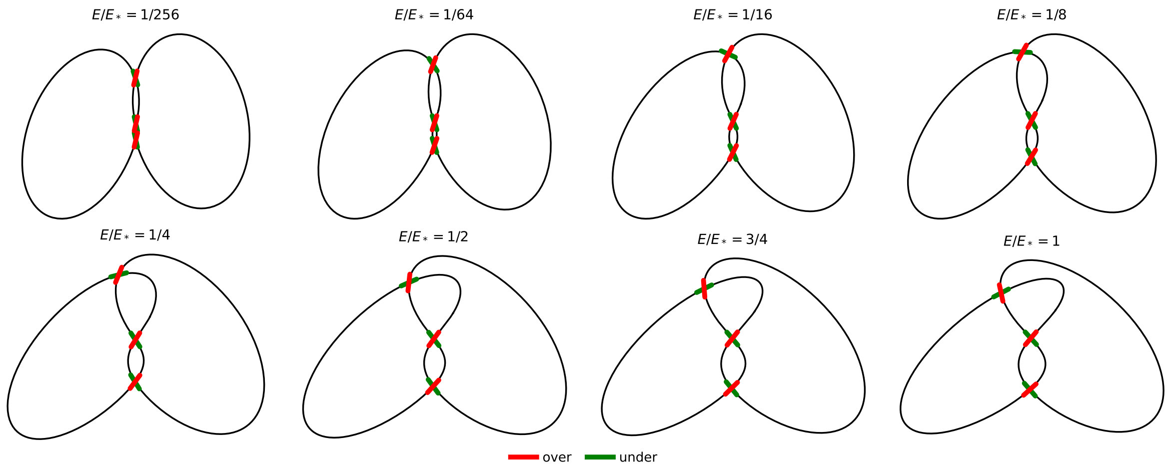

Computed family. Complete stereographic images of K_E, shape-normalized by $R = \sqrt{E}$, with one fixed screen view. Short green/under and red/over segments are centered at the computed crossings and red is always drawn last. The range ends at the first positive ODE singularity $E_* = 4/27$, not at a knot transition.

Model $C = \{z^2 = w^3\} \subset (\mathbb{C}^2, \omega_0), \quad \omega_0 = \frac{i}{2}(dz \wedge d\bar{z} + dw \wedge d\bar{w}), \quad H = |z|^2 + |w|^2, \quad K_E = C \cap H^{-1}(E).$

Parametrization $z = s^3 = u^{3/2} e^{3i\theta}, \quad w = s^2 = u e^{2i\theta}, \quad E = h(u) = u^3 + u^2 = u^2(u+1), \quad u > 0.$

Restricted forms $\lambda_0 = \frac{1}{2}\,\mathrm{Im}(\bar{z}\,dz + \bar{w}\,dw), \quad \lambda_C = \frac{1}{2}(3u^3 + 2u^2)d\theta, \quad \omega_C = d\lambda_C = \frac{1}{2}(9u^2 + 4u)du \wedge d\theta.$

Time form $dh = (3u^2 + 2u)du, \quad \omega_C = dh \wedge \eta, \quad \eta = \dfrac{9u+4}{2(3u+2)}d\theta, \quad \iota_{X_h}\omega_C = -dh,\ \eta(X_h) = 1.$

Period / action $T(E) = \displaystyle\oint_{K_E} \eta = \pi\frac{9u+4}{3u+2}, \qquad A(E) = \oint_{K_E} \lambda_C = \pi(3u^3 + 2u^2), \qquad \frac{dA}{dE} = T(E).$

Energy-native function $\Phi(E) = 3 - \dfrac{T(E)}{\pi} = \dfrac{2}{3u+2}, \qquad u = \dfrac{2(1-\Phi)}{3\Phi}.$

Algebraic elimination Substituting $E = u^2(u+1)$ gives $(4 - 27E)\Phi^3 - 12\Phi + 8 = 0$.

Direct derivatives $\Phi_E = -\dfrac{6}{u(3u+2)^3}, \qquad \Phi_{EE} = \dfrac{12(6u+1)}{u^3(3u+2)^5}.$

ODE $E(27E-4)\Phi'' + 2(27E-1)\Phi' + 6\Phi = 0.$

Gauss form With $x = \dfrac{27E}{4}$ and $\Phi(E) = y(x)$, $x(1-x)y'' + \left(\dfrac{1}{2} - 2x\right)y' - \dfrac{2}{9}y = 0.$

Solution $\Phi(E) = {}_2F_1\!\left(\dfrac{1}{3}, \dfrac{2}{3}; \dfrac{1}{2}; \dfrac{27E}{4}\right) - \dfrac{3}{2}\sqrt{E}\,{}_2F_1\!\left(\dfrac{5}{6}, \dfrac{7}{6}; \dfrac{3}{2}; \dfrac{27E}{4}\right).$

Branch sign For $E > 0$, $u > 0$ and hence $\Phi(E) = 2/(3u+2) < 1$; therefore $\Phi(E) = 1 - \dfrac{3}{2}\sqrt{E} + O(E)$ and the minus sign is forced.

Integral uniformizer Only now set $q = \dfrac{\sqrt{E}}{4}$ and $G(q) = \Phi(16q^2)$. Then

$$G(q) = 1 - 6q + 48q^2 - 420q^3 + 3840q^4 - 36036q^5 + 344064q^6 - 3325608q^7 + \cdots.$$

Coefficients $G(q) = \displaystyle\sum_{n\geq 0}(-1)^n a_n q^n, \qquad a_n = 4^n\binom{3n/2}{n} \in \mathbb{Z}, \qquad a_n = [x^n]\big((1-x)(1-2x)\big)^{-n-1}.$

Recurrence / OEIS $(n+1)(n+2)a_{n+2} = 12(3n+4)(3n+2)a_n, \quad a_0 = 1,\ a_1 = 6.$ Unsigned coefficients: **OEIS A244038**.

First terms $1,\ 6,\ 48,\ 420,\ 3840,\ 36036,\ 344064,\ 3325608,\ 32440320,\ 318704100, \ldots$

Numerical audit Curved radial-ring quadrature on the actual surface in $\mathbb{R}^4$, using finite-difference tangents and no analytic pullback density, agrees with $A(E)$ to 1.1×10^{-11} relative and recovers $T = dA/dE$ to 8.3×10^{-11}. Independent constrained-flow quadrature also agrees to 8.3×10^{-11}.

Embedded verification payload. Attachments: exact verifier, constrained-flow audit, curved-area audit, flat-mesh audit, crossing data, figure code, JSON payload, and this LaTeX source.